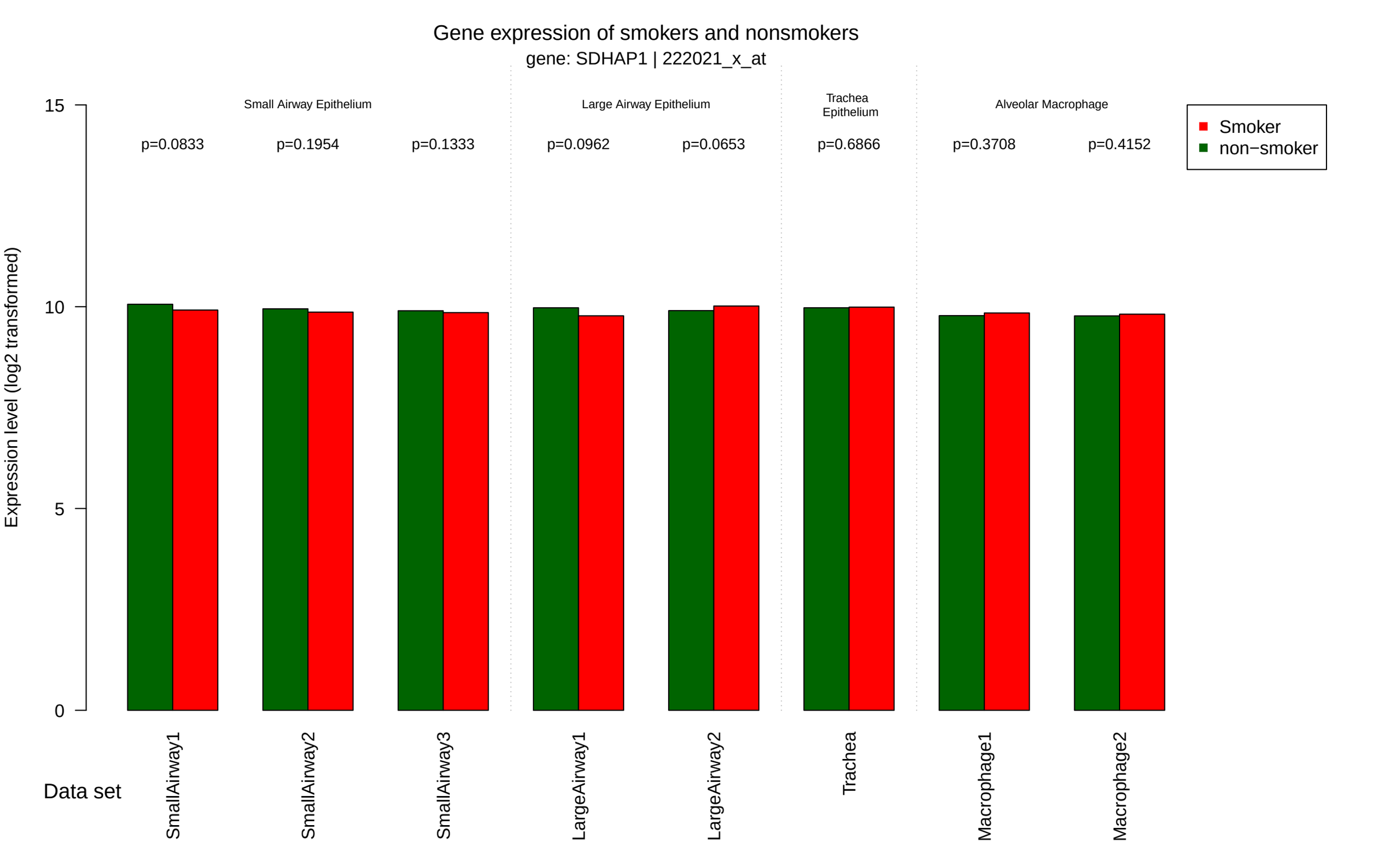
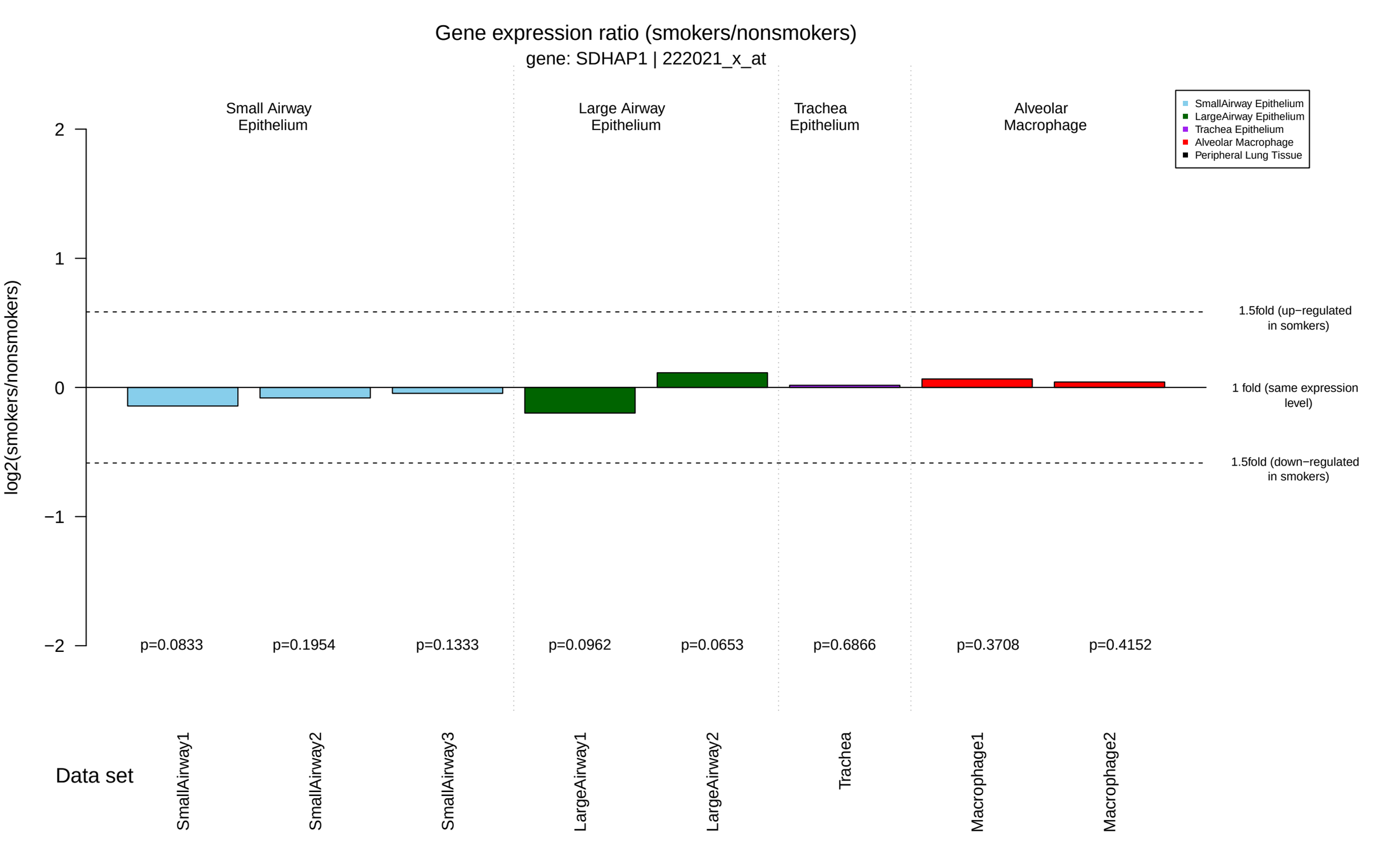
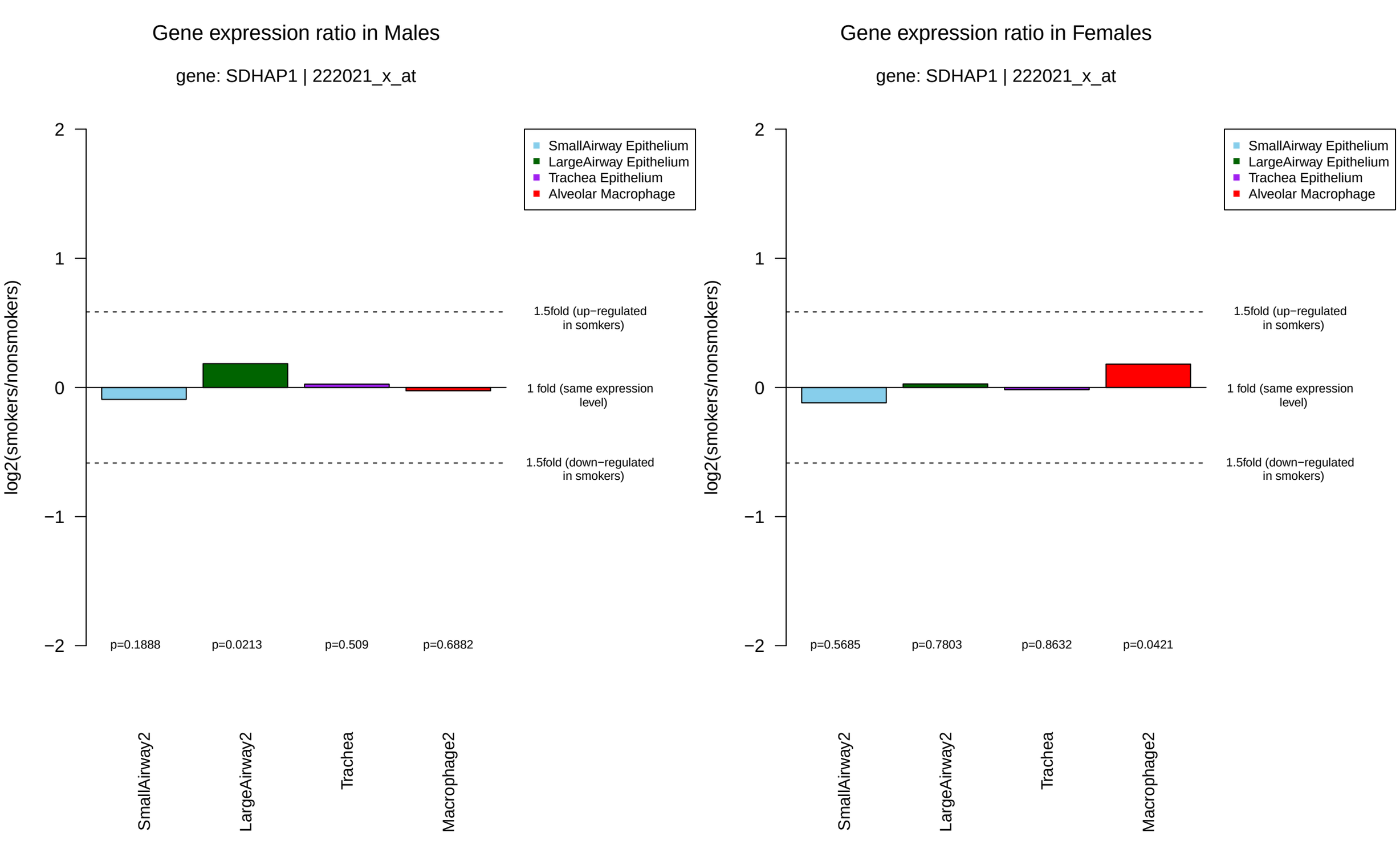
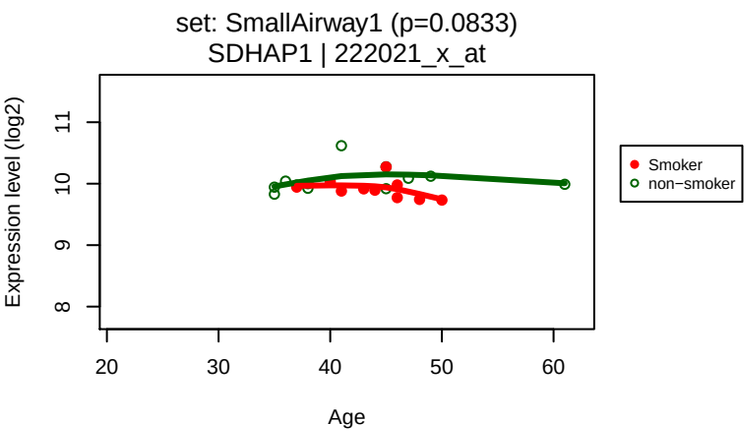
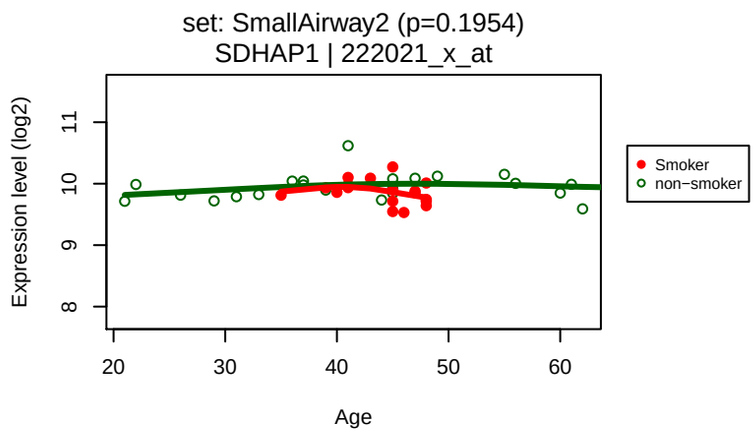
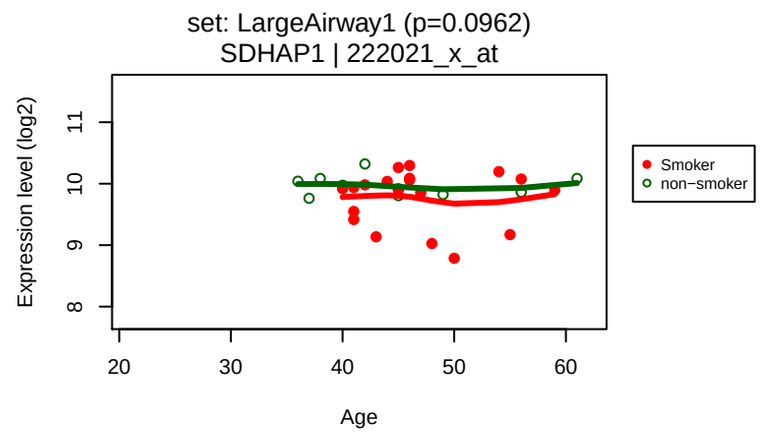

Gene expression of smokers and nonsmokers
gene: SDHAP1 | 222021_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0833
p=0.1954
p=0.1333
p=0.0962
p=0.0653
p=0.6866
p=0.3708
p=0.4152
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

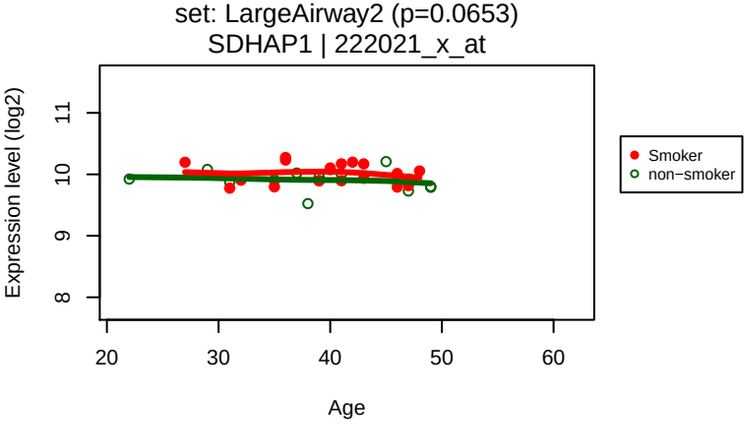

Gene expression ratio (smokers/nonsmokers)
gene: SDHAP1 | 222021_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0833
p=0.1954
p=0.1333
p=0.0962
p=0.0653
p=0.6866
p=0.3708
p=0.4152
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

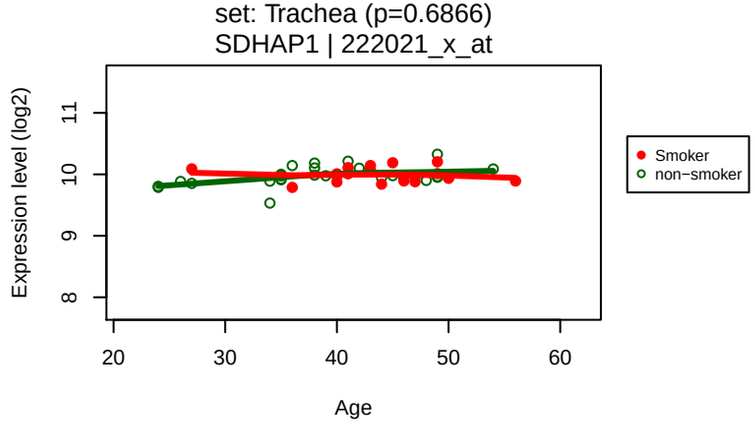

Gene expression ratio in Males
gene: SDHAP1 | 222021_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1888
p=0.0213
p=0.509
p=0.6882
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SDHAP1 | 222021_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5685
p=0.7803
p=0.8632
p=0.0421
SmallAirway2
LargeAirway2
Trachea
Macrophage2

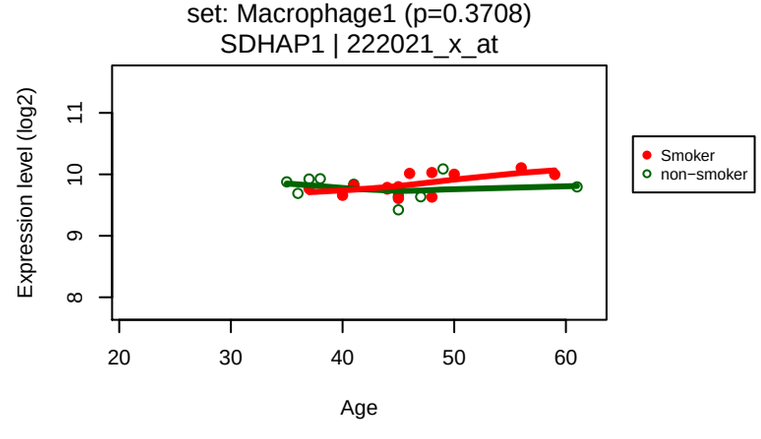

set: SmallAirway1 (p=0.0833)
SDHAP1 | 222021_x_at
Expression level (log2)
Age
Smoker
non−smoker

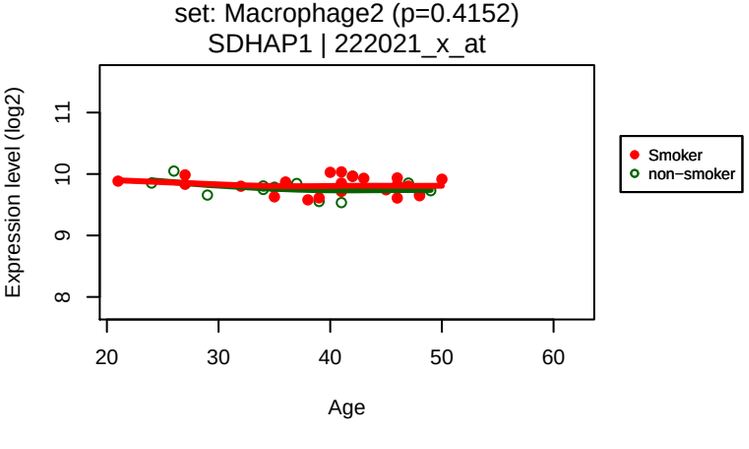

set: SmallAirway2 (p=0.1954)
SDHAP1 | 222021_x_at
Expression level (log2)
Age
Smoker
non−smoker

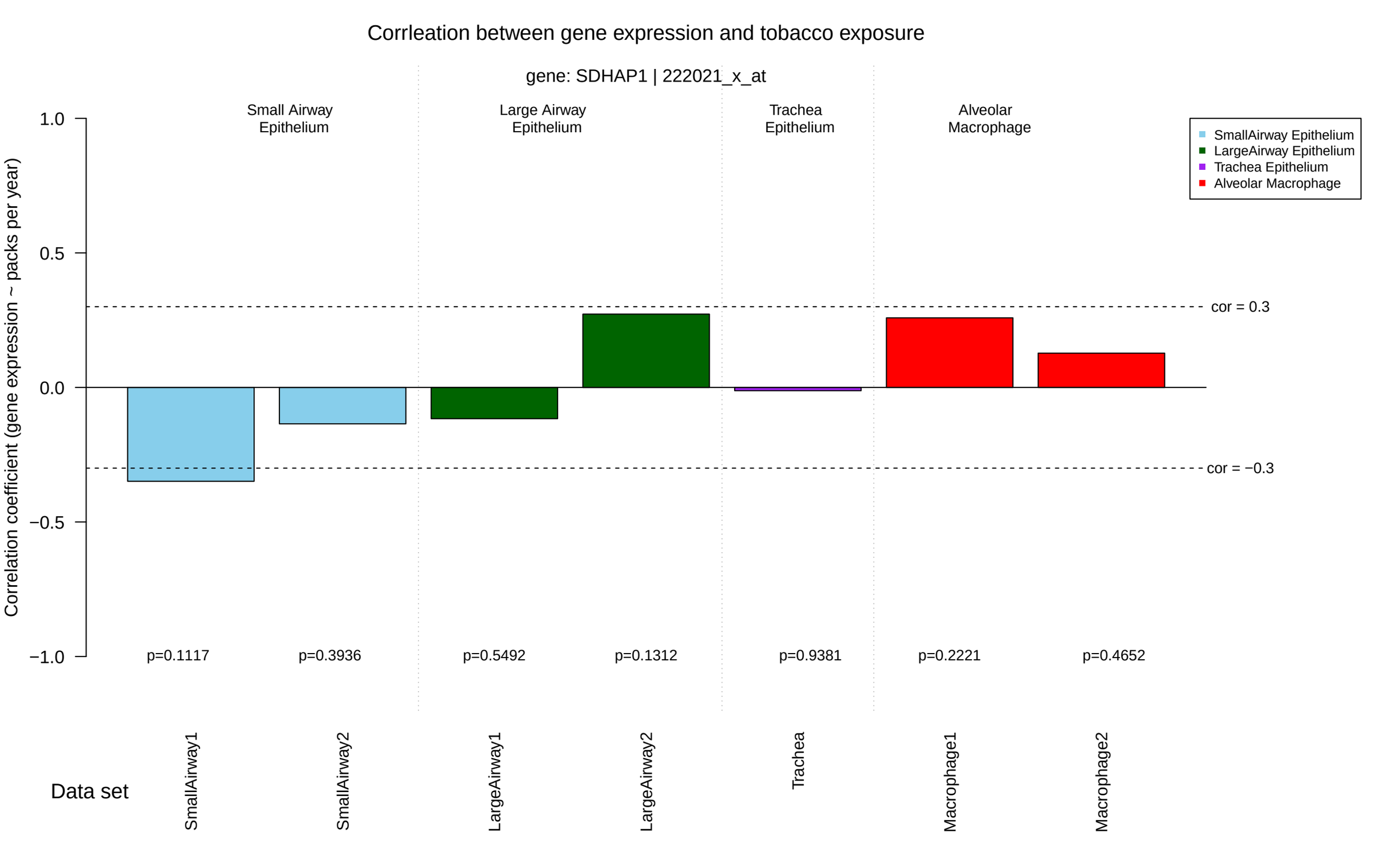

set: LargeAirway1 (p=0.0962)
SDHAP1 | 222021_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0653)
SDHAP1 | 222021_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6866)
SDHAP1 | 222021_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3708)
SDHAP1 | 222021_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4152)
SDHAP1 | 222021_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SDHAP1 | 222021_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1117
p=0.3936
p=0.5492
p=0.1312
p=0.9381
p=0.2221
p=0.4652
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2